

- Its deficiency destroys the muscles and causes abnormal functioning of the reproductive system in males as well as in females.

Vitamin-K (Phylloquinone)

- It was discovered by **Henrik Dam** (1935).
- It is also called as **naphthoquinone** and is synthesised in body by some bacteria.
- It is a coagulation vitamin, that is why it helps in clotting of blood.
- The main sources are cauliflower, spinach, tomatoes, soyabean, etc.
- Its deficiency delays clotting of blood and causes haemorrhage that is why also called **antihæmorrhagic vitamin**.

Water Soluble Vitamins

These are the vitamins, which are soluble in water that is why excess of these vitamins are normally excreted through the urine.

Vitamin-B₁ (Thiamine)

- The name of vitamin-B₁ was given by **Funk** and discovered by **Eijkman**.
- It is destroyed on cooking and get dissolved in cooking water.
- It is helpful in normal functioning of nerve cells and metabolism.
- It also helps in maintaining normal appetite and digestion.
- Sources are yeast, rice, wheat, grains, beans, soyabean, liver oil, milk, etc.
- Its deficiency causes beri-beri (loss of appetite, paralysis of legs and head, etc).

Vitamin-B₂ (Riboflavin)

- It was first isolated by **Warsung** and **Christian** from animal tissue.
- These are destroyed on cooking and in strong sunlight. It helps in protein and fat metabolism.
- It is necessary for normal growth of body.
- The main sources of vitamins are milk, eggs, liver, green leafy vegetables, pulses, cheese, etc.
- Its deficiency causes inflammation of tongue, cornea, cracks appear at the corner of the mouth.

Vitamin-B₃

(Niacin or Nicotinic Acid)

- It is also known as antipellagra factor.
- It helps in oxidation of carbohydrates, proteins and fats.
- The main sources are cereals, liver, maize, fruits, milk, egg, meat, yeast, kidney, etc.
- Its deficiency causes pellagra, dermatitis, diarrhoea.

Vitamin-B₅ (Pantothenic Acid)

- It is a part of coenzyme-A required for cell respiration, normal nerves and skin.
- Its sources include yeast, peas, liver, eggs, kidney, etc.
- Its deficiency can lead to anaemia, dermatitis and nerve degeneration.

Vitamin-B₆ (Pyridoxine or Pyridoxal)

- It is an essential part of coenzymes for amino acid synthesis.
- Extreme excess of this vitamin leads to poor control and coordination.
- The main sources include meat, milk, liver, eggs, etc.
- Its deficiency can lead to disturbance of central nervous system and anaemia.

Vitamin-B₇ (Biotin)

- It is a sulphur containing substance also called **vitamin-H**.
- It serves as coenzyme for fatty acid synthesis.
- Sources include fruits, vegetables, liver, milk, eggs, etc.
- Its deficiency can lead to scaly and itchy skin, muscle pain and weakness.

Vitamin-B₉ (Folic Acid)

- It is also called **vitamin-M**.
- It was first isolated from the spinach leaves.
- It is essential for growth and maturation of red blood cells.
- This vitamin plays a vital role in various metabolic disorders. Some main sources are green leafy vegetables, yeast, banana, pulses, cauliflower, meat, liver, etc.
- Its deficiency causes macrocytic anaemia in man.

Vitamin-B₁₂ (Cyanocobalamine)

- It is a dark coloured cobalt based pigment, also known as **cobamide cyanide**.
- It helps in RBC formation and proper functioning of nerve tissues.
- The main sources are liver, cheese, milk, meat, fish, eggs, kidney, etc.
- Its deficiency can lead to **pernicious anaemia**.
- It is not found in plants, however it has been considered that alga and *Spirulina* contain this vitamin.

Vitamin-C (Ascorbic Acid)

It is an earliest known vitamin, which is essential for keeping teeth, gums and body joints healthy.

- It prevents the formation of stones in gall-bladder.
- It is quickly and easily destroyed by heat and increases resistance of our body against infections, i.e. viral infections and common cold.
- The main sources are amla, *Citrus* fruits (lemon and orange), guava, tomatoes, peppers, etc.
- Its deficiency can lead to scurvy (bleeding and loosening of gums).

Water

It is an inorganic compound made up of H and O in the ratio of 2 : 1.

- Human body contains about 65% water.
- It regulates body temperature by sweating and evaporation.
- It helps in digestion, transportation and excreting body wastes.
- The main sources are metabolic water, liquid food and drinking water.
- Abnormal depletion of body fluid leads to dehydration (loss of water).

Roughage

It is fibrous material present in plants.

- It does not take part in growth as we cannot digest it.
- It adds bulk in food item, to prevent constipation and helps in retaining water in body and aids in proper working of digestive system.
- The main sources are salads, outer layers of grains cereals vegetables and porridge (dalia).

Balanced Diet

A diet Which contains all the essential nutrients in required quantity.

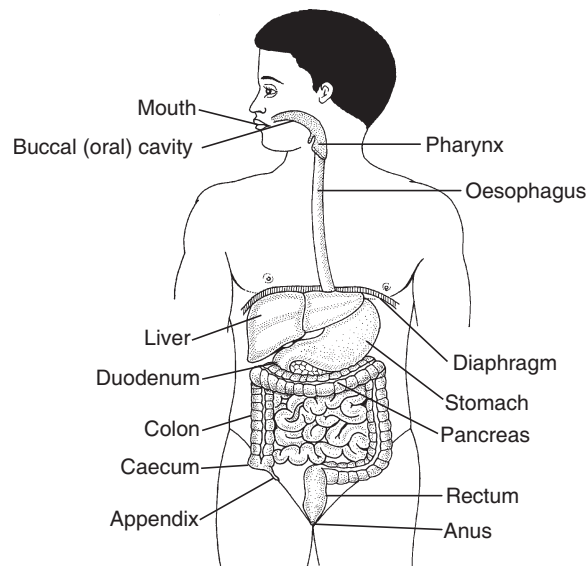
- It is related to state of one's age, health and occupation.
- Body requires carbohydrates, proteins and fats in the approximate ratio of 4 : 1 : 1 for good health.
- Growing child needs more protein than a grown up man. Person doing physical work (athlete, carpenter, rickshaw puller, etc.) need more carbohydrates and fats.
- A pregnant woman needs more proteins, calcium, potassium in her diet.
- Person recovering from illness need more proteins, minerals and vitamins.
- The eating of a particular type of diet due to the food habit leads to deficiency diseases.

DIGESTIVE SYSTEM

Parts of body concerned with digestion, absorption and assimilation of food followed by elimination of undigestible remains is called **digestive system**.

- Digestion involves splitting of food molecules by hydrolysis into smaller molecules that can be absorbed through the epithelium of the gastrointestinal tract.

- Man and other animals have holozoic nutrition, i.e. solid form of food.



Structure of digestive system

Dentition

The oral or buccal cavity of vertebrates contain masticatory organs called **teeth** or **dentes**. Mammal contains two sets of teeth. The first set is called **milk teeth** and second set is called **permanent teeth**.

- There are 32 permanent teeth in man.
- These are of following four type:
 - (i) Incisors (for cutting) four in numbers.
 - (ii) Canines (for tearing) two in numbers.
 - (iii) Premolars (for grinding) four in numbers.
 - (iv) Molars or wisdom teeth (for grinding) six in numbers.
- In elephants, the tusks are the incisors of upper jaw.
- Maximum number of teeth are present in **horse** and **pig**.
- Hardest part in the body is tooth enamel.
- Main bulk of tooth is formed of dentine.

Dental Formula of some Common Mammals

Mammal	Dental formula	Total number of teeth
Man (child)	2102/2102	20
Man (adult)	2123/2123	32
Horse	3143/3143	44
Dog	3142/3143	42
Cow and sheep	0033/3133	32
Cat	3121/3121	30
Rabbit	2033/1023	28
Mouse	1003/1003	16

Digestion

The food we intake is in the form of biomacromolecules and thus, cannot be utilised by our body. The complex food substances are converted into simpler absorbable form in the process called **digestion**.

Digestion takes place in following five steps:

- (i) Ingestion of food (ii) Digestion of food
- (iii) Absorption of digested food
- (iv) Assimilation and egestion of unwanted food

Ingestion of Food

Ingestion takes place in buccal cavity. Food is taken through mouth cavity. Salivary glands lubricate the food and bind the food particles together to form bolus. Salivary glands have starch splitting enzyme called **ptyalin**. Food is masticated by teeth and swallowed.

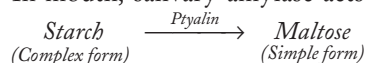
Digestion of Food

Process of converting complex, insoluble food particles into simple soluble and absorbable form is called digestion. It takes place in various parts of digestive system.

Digestion in Mouth

Salivary glands secrete mucus to moist the food and to digest starch.

- In mouth, salivary amylase acts on starch.



- pH of saliva = 6-7.4

Digestion in Stomach

The food passes down through the oesophagus into stomach. Now, food is mixed with gastric juice and hydrochloric acid, which disinfect the food and create acidic medium.

- The pH of gastric juice is 4.5.
- Pepsin digests proteins and converts them into peptones and renin converts milk to curd. Digested food now is called **chyme**.
- The opening of stomach into small intestine is called **pylorus**. Alcohol is easily diffusible across the plasma membrane.

Digestion in Small Intestine

Now, chyme moves to duodenum. In small intestine, food receives three alkaline secretions—bile from liver, pancreatic juice from pancreas and intestinal juice from intestinal glands.

- (i) **Bile** Food is mixed with bile (liver) to breakdown fats into smaller globules.
- (ii) **Pancreatic Juice** It contains following three enzymes:
 - (a) **Trypsin** Which acts upon proteins and breaks them into peptides.
 - (b) **Amylase** Which converts starch into simple sugar.
 - (c) **Lipase** Which converts fats into fatty acids and glycerol.

- (iii) **Intestinal Juice** Food passes into ileum and mixes with intestinal juice. It contains following three types of enzymes:

- (a) **Maltase** Which converts maltose into glucose.
- (b) **Lactase** Which converts lactose into glucose.
- (c) **Sucrase** Which converts sucrose into glucose.
- Food is now called **chyle**.

Absorption and Assimilation of Digested Food

Ileum's internal surface has finger-like folds called **villi**. There is a dense network of blood capillaries and lymph capillaries in each villus. It helps in absorption of food.

- Absorption occurs by two types of processes, i.e. **passive** and **active absorption**.
- Passive absorption occurs down the concentration gradient and **active absorption** is independent of concentration gradient.
- Assimilation is followed thereafter, i.e. the absorbed food materials are incorporated into tissue cells.

Egestion of Unwanted Food

- After complete assimilation digested food passes into large intestine. Large intestine cannot absorb food, but absorbs much of the water. The remaining semi-solid waste is called **faeces** and is passed into rectum. It is expelled out through anus.
- Sometimes, the acids in stomach increases causing indigestion to get rid of the stomach aches milk of magnesia, a base is taken to neutralise the excess acid in stomach.

RESPIRATION

Respiration is an oxidising and energy liberating process, during which complex organic compounds are broken down into simpler substances. It is brought about by respiratory organs.

Respiratory organs are different in different animals:

- (i) **Lungs** In man, frog, birds, lizards, camel and cattles, etc.
- (ii) **Skin** In frog, earthworm.
- (iii) **Gills** In fishes, prawn.
- (iv) **Trachea** In insects.

General body surface In *Amoeba*, *Euglena*, *Spirogyra* and *Chlamydomonas*.

Types of Respiration

Respiration is of two types based on the presence or absence of O_2 .

1. Aerobic Respiration

It is the process, in which respiratory substrates are oxidised into CO_2 , water and energy, in the presence of oxygen. This is common in higher organisms.



One molecule of glucose is converted into two molecules of pyruvic acid, the process is known as **glycolysis**. It takes place in cytoplasm of cells.

- Pyruvic acid releases energy + CO_2 + H_2O (Krebs' cycle).
- 1 molecule of glucose = 36 ATPs in eukaryotic cell.
- 1 molecule of glucose = 38 ATPs in prokaryotic cell.

2. Anaerobic Respiration

It is the process, in which the respiratory substrates are incompletely oxidised into CO_2 and alcohol.

It occurs in the absence of oxygen. It is also called as **fermentation**.



It occurs in lower organisms such as bacteria and fungi and in higher organisms under the condition when O_2 is limiting.

Respiratory Quotient (RQ)

The ratio of volume of CO_2 evolved to volume of O_2 consumed in respiration is called RQ.

$$RQ = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$$

Respiratory substrate	RQ
Carbohydrates	1
Fats	0.7
Organic acids	4

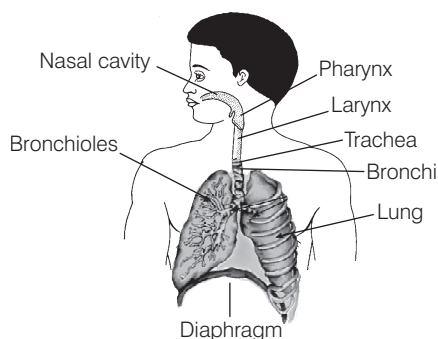
Respiration in Human

Respiration in humans takes place in two phases:

External Respiration (Breathing)

It comprises of inspiration and expiration of air.

- (i) **Inspiration** This is the phenomenon, in which intake of air takes place. Muscles of the diaphragm contract and diaphragm flattens. The lower ribs are raised upward and outwards. The chest cavity enlarges, the air pressure in the lungs is decreased, air rushes into the lungs.



Human respiratory system

- (ii) **Expiration** This is the phenomenon, in which air is exhaled out.

- Relaxation of muscles of the ribs and diaphragm takes place. Diaphragm again becomes dome-shaped.
- Chest cavity is reduced and air is forced outward through nose and trachea.
- Normal breathing rate in man is about 18-20 per min.

Internal Respiration

It is a complex process, in which food is broken down to release energy.

- Biochemical phase takes place inside the cell.
- Overall passage of air in human is as follows:
 - Nostrils → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli → Cells → Blood.

CIRCULATORY SYSTEM

Circulation of body fluids is also referred to as **circulatory system** or **internal transport**.

In higher and multicellular organisms, there is no direct supply of the useful materials and the removal of wastes from the body cells, so they need a transport system called **circulatory system**. Circulation of nutrients and hormones takes place in different directions.

- It transports nutrients, i.e. glucose, fatty acids, vitamins, etc., from the site of absorption to the different parts of the body.
- It transports nitrogenous wastes, oxygen, hormones, water, chemical substances, etc., from different parts of the body to organ tissue/cells of direction.

Types of Circulatory System

Blood circulatory system is of two types:

1. Open Circulatory System

Blood may be present in blood vessels for sometime, but finally comes out from the blood vessels in open spaces called **sinuses**. It is found in leeches, among the annelids, cockroach, prawns, insects, spiders, starfish, etc.

- Blood flows with very slow velocity and at low pressure.
- In cockroach, blood circulation is completed in 5-6 minutes.
- Blood is in direct contact with the tissues.
- Respiratory pigment if present is dissolved in plasma.

2. Closed Circulatory System

Blood flow in closed vessels. It is found in earthworm, *Nereis*, molluscs and all vertebrates. Blood flows with high speed and at high pressure.

- Exchange of materials occurs through the tissue fluids.
- Blood does not come in direct contact with tissue cells. Respiratory pigment is present in RBCs.

Blood Vascular System

Blood vascular system consists of three components:

1. Blood Vessels

These are elastic muscular tubes that carry blood.

These are:

- (i) **Arteries** These are the thick-walled blood vessels, which carry the blood away from the heart to various body parts.
They are deep seated in body and have no valves in them and carry oxygenated blood in them, except the pulmonary artery, which carries deoxygenated blood to the lungs. Blood flows at high pressure and high speed.
- (ii) **Veins** These are thin-walled blood vessels, which carry blood from various body parts to the heart. Superficial in position and have valves in them to prevent back flow of blood in them, blood flows at low pressure and lower speed. They carry deoxygenated blood in them, except the pulmonary vein, which carries oxygenated blood to the heart.
- (iii) **Capillaries** These are the thinnest blood vessels, which connect arteries to the veins. Each capillary is lined by a single layer of flat cells. These help in exchange of materials like the nutrients, gases, waste products, etc., between blood and cells.

2. Blood

Blood is a red vascular connective tissue. An average adult person has about 4-6L of blood. It forms 6-10% of body weight and 30-35% of extracellular fluid. Blood has specific gravity of 1.06. It is a viscous fluid due to the presence of formed elements.

It has a saltish taste and is mildly alkaline with a pH = 7.4. The alkaline nature of blood contributes to the oxygen carrying capacity of haemoglobin as in acidic pH this is decreased. It is bright red when oxygenated and purple when deoxygenated. The study of blood is called **haematology**. People at higher altitudes have more blood than those at lower altitudes. This extra blood supplies the additional oxygen to body cells. It is sticky and opaque fluid with a viscosity of 4.7.

The blood is formed of two parts:

- (i) Plasma
- (ii) Blood corpuscles

(i) Plasma

Plasma is faint yellow-coloured non-living fluid present in blood. It is slightly alkaline and viscous. It constitutes about 55-60% of the blood volume. It is composed of following components in the table below:

Components of Plasma	
Water	90-92%
Inorganic salt	1-2%
Plasma protein in colloidal state	6-7%
Other organic compounds	1-2%

Functions of Plasma

- It transports simple food components (glucose, amino acids, etc.) from intestine and liver to other body parts.
- It transports metabolic wastes like urea, uric acid, etc., from body tissues to the kidneys or their excretion.
- It transports hormones from endocrine glands to the target organs.
- It helps in keeping a constant pH of blood.
- Blood proteins and fibrinogen, present in plasma, help in blood clotting at the injury. It forms the tissue fluid, which keeps the tissue moist and helps in exchange of material in between the blood and cells.

(ii) Blood Corpuscles or Blood Cells

Blood contains certain floating bodies called **formed elements**. These floating elements include blood corpuscles or blood cells. These form 40-45% of blood.

- The percentage of blood cells is called **haematocrit value** of packed cell volume.
- They are of three types:
 - (i) Red Blood Corpuscles (RBCs) or **Erythrocytes**
 - (ii) White Blood Corpuscles (WBCs) or **Leucocytes**
 - (iii) Blood platelets

Red Blood Corpuscles (RBCs)

Shape in all vertebrates except mammals, these are oval, biconvex, nucleated and non-nucleated.

- In mammals except camel and llama, the mature RBCs are circular, biconcave and non-nucleated.
- **Number** The RBCs are much more in number than WBCs. Normal RBC count is slightly lower in a woman (i.e. 4500000/cu mm) than a man (i.e. 5000000/ cu mm).
- The instrument used to determine RBC count is **haemocytometer**.
- Thrombocytes, also called **spindle cell**. These are spindle-shaped and found in blood of vertebrates other than mammals. These have oval nucleus and help in blood clotting in vertebrates.
- Physiological state and high altitudes increase the RBC count. RBC count decreases due to haemorrhage, haemolysis, etc., and is called **anaemia**. If RBC increases much more than the normal level, it is called **polycythemia**.
- **Lifespan** Average lifespan of human RBC is 120 days, while it is 100 days in frog and 50-70 days in rabbit.
- **Haemoglobin** (*Haem*-iron containing pigment, *globin* - protein). One RBC has about 280 haemoglobin molecules.
- **Structure** A red blood cell is bounded by elastic semipermeable membrane. It does not possess cell organelles. This feature enables it to hold more haemoglobin due to availability of space.



Structure of erythrocytes